

Composition Pair B1.15 + A1.10 — Determinism Makes Action Layer Evidence Reliable for Action-Feedback, and Action-Feedback Gives Determinism Its Governance Significance

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This work derives from and formalizes the instinct/reasoning separation pattern introduced in “The Instinct/Reasoning Separation Outside the Model” (Li, April 2026), the second paper in the CKS theory series following “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems” (Li, April 2026).

This note formalizes composition pair B4.27 in the Phase B4 series, identifying the enabling relationship between B1.15 (multi-shaped human governance across evolution mechanisms, specifically the action-feedback governance pathway) and A1.10 (the determinism contract). It does not introduce new axioms. Every claim is traceable to the source papers or to prior derivation notes in this series.

Abstract

Two architectural commitments in the CKS theory series — the multi-shaped governance of action-feedback evolution (B1.15, Paper 2 §8) and the determinism contract (A1.10, Paper 1 §4.1, §6.2, §11.3) — stand in an enabling relationship with a specific structure. A1.10 determinism enables B1.15’s action-feedback governance pathway by ensuring that Action layer records are interpretable as behavioral pattern evidence rather than noise: when behavior is deterministic, records of outputs reliably reflect what the entity’s current DNA specification produces for given inputs, making those records the raw material for evidence-based improvement proposals under human-mediated governance. Without determinism, Action layer records are an unreliable mixture of DNA-governed patterns and nondeterministic variation that proposing substrates cannot disentangle. In the reverse significance direction, B1.15 gives A1.10 governance significance beyond operational correctness: determinism enables the deployment’s operational experience to function as a continuous evidence base for governed improvement, and without the action-feedback pathway to consume that evidence, determinism’s governance contribution remains static and limited to reproducibility verification. The pair’s co-presence requirement is strict: action-feedback without determinism evaluates noise; determinism without action-feedback accumulates reliable evidence that is never used for improvement. The primary threat to the pair is mutation per B1.13 — LLM updates that may introduce nondeterminism or shift behavioral

patterns — which is why mutation governance instruments protecting A1.10 are directly relevant to B1.15's evidence reliability.

1. Pair Identification

Pair: B1.15 (multi-shaped human governance across evolution mechanisms, specifically the action-feedback governance pathway; Paper 2 §8) and A1.10 (the determinism contract; Paper 1 §4.1, §6.2, §11.3).

Relationship type: Enabling pair. The pair has an asymmetric primary enabling direction — A1.10 enables B1.15 with evidence reliability — and a reverse significance direction in which B1.15 gives A1.10 its governance significance as evidence foundation for continuous improvement.

Supporting references:

- B2.26 (Action layer) — the evidence substrate whose reliability depends on A1.10.
- B2.74 (proposing substrates) — the entities that evaluate Action layer records to form improvement proposals.
- B2.75 (Stage 1 and Stage 2 governance) — the human-mediated approval process that determines whether proposals become DNA changes.
- B2.76 (action-feedback proposals) — reliable only when A1.10 holds.
- B1.14 (directed selection) — the mechanism that translates evidence-informed proposals into DNA changes.
- B1.13 (instinct evolution as undirected mutation) — the primary operational threat to A1.10 and thereby to the pair.
- A5.06 (cell behavior determinism test) and A5.16 (reproducibility test) — the operational verifications that A1.10 is satisfied.
- B3.28 (Evidence Blindness) — the anti-pattern that materializes when either commitment is absent.

2. Enabling Direction: A1.10 Provides the Evidence Reliability That B1.15 Requires

The action-feedback governance pathway specified in B1.15 operates over evidence accumulated in the Action layer per B2.26. The evidence consists of records of operations: what inputs were received, what outputs were produced, what sequences occurred over time. Proposing substrates per B2.74 evaluate this evidence to identify behavioral patterns and form proposals for DNA changes that would improve the behavioral tendencies the entity exhibits. Stage 1 and Stage 2 governance per B2.75 evaluate those proposals, and authorized proposals become DNA changes through directed selection per B1.14.

The entire pathway depends on a property that is not internal to B1.15: the Action layer records must *reliably reflect the behavioral pattern the entity's current DNA specification produces*. If they do not, the proposing-substrate evaluation is operating on noise rather than signal, and proposals will describe patterns that may have no stable relationship to what the current DNA actually produces.

A1.10 determinism is the property that establishes this reliability. The determinism contract specifies that given the same DNA specifications and the same inputs, the same outputs are produced. Under A1.10, if the Action layer shows that input type X consistently produced output Y during the period under evaluation, then output Y is what the current DNA specification produces for input type X — not an artifact of a nondeterministic process that happened to produce Y on those occasions. The record is interpretable as behavioral pattern evidence.

The contrast with nondeterminism makes the enabling relationship precise. If behavior were nondeterministic — if the same DNA specification and the same inputs sometimes produced Y and sometimes produced Z — then Action layer records would be an unreliable mixture: some records reflecting the DNA’s deterministic contribution and some reflecting variation the DNA does not uniquely determine. Proposing substrates per B2.74 would be unable to distinguish “the current DNA consistently produces Y for X” from “Y appeared more frequently than Z in this sample due to nondeterministic variation.” Proposals built on this ambiguity would describe apparent patterns that may not persist in the DNA’s ongoing behavior.

A1.10 collapses this ambiguity. Under the determinism contract, patterns in Action layer records are DNA patterns, not sampling artifacts. The proposing substrate’s task becomes well-defined: find stable input-output patterns in the records and propose DNA changes that would improve them. A1.10 is therefore a prerequisite for B1.15’s action-feedback governance pathway to function as intended — not as a best-effort correlation finder, but as an evidence-based interpreter of DNA-governed behavioral tendencies.

3. Significance Direction: B1.15 Activates A1.10’s Governance Contribution

A1.10 determinism has operational significance in its own right. The five guarantees of the determinism contract make deployments reliable, reproducible, and testable. A5.06 and A5.16 verify the contract in operation. This operational significance does not depend on B1.15.

But A1.10’s significance for *governance* — its contribution to the deployment’s capacity to learn from its operational experience and improve under human oversight — is only realized through the action-feedback pathway that B1.15 specifies. Without B1.15, the reliable evidence that determinism produces in the Action layer accumulates and is never systematically converted into governed improvement proposals. A5.16 reproducibility verification confirms that determinism holds; static confirmation that behavior is reproducible is a different contribution from the continuous evidence-based improvement architecture that action-feedback enables.

B1.15 activates A1.10’s governance significance by providing the pathway through which reliable Action layer evidence becomes governed improvement proposals, and through which governed improvement proposals become DNA changes under directed selection per B1.14. The deployment that has A1.10 and B1.15 together has both reliable evidence production (A1.10) and a governed channel through which that evidence drives improvement (B1.15). The deployment that has A1.10 alone has reliable evidence that is not consumed for improvement. Determinism’s governance contribution — its

role in making the deployment's operational history an evidence base for continuous, governed improvement — is a function of whether B1.15 is present to receive and process that evidence.

4. The Evidence Quality Chain and the Cost of Breaking It

The enabling relationship between A1.10 and B1.15 produces a chain of dependent steps. Each step inherits its reliability from the step that precedes it.

- **A1.10 determinism → reliable Action layer records per B2.26.** Under the determinism contract, records of outputs reliably reflect what the current DNA specification produces for given inputs. This is the foundational guarantee. Without it, no downstream step can compensate.
- **Reliable records → reliable proposing-substrate evaluation per B2.74.** Proposing substrates evaluate DNA-governed behavioral patterns, not sampling artifacts. Candidate proposals inherit the reliability of the evidence they summarize.
- **Reliable proposals per B2.76 → sound governance evaluation per B2.75.** Stage 1 and Stage 2 governance evaluate proposals that accurately describe behavioral tendencies the current DNA exhibits. The governance decision is made on a sound evidence basis.
- **Sound governance → evidence-informed directed selection per B1.14.** Authorized DNA changes address behavioral patterns that the evidence accurately characterized. The improvement targets what the evidence described.
- **Improved DNA → more informative future Action layer records.** Each pass through the cycle refines the deployment's behavioral patterns and produces richer evidence for the next round of action-feedback.

Breaking A1.10 at the chain's foundation propagates unreliability through every downstream step. Proposing substrates evaluate noise; proposals describe apparent patterns that may not persist; governance authorizes changes based on inaccurate behavioral characterizations; directed selection modifies DNA in ways that may not address the tendencies that motivated the change. The improvement cycle degrades not at one point but at every point that depends on Action layer evidence.

This chain structure is what makes A1.10 an *enabling* condition for B1.15's action-feedback pathway rather than a *contributing* condition. A contributing condition can be partially compensated by improvements elsewhere. An enabling condition cannot: without A1.10, the chain has no foundation, and no amount of sophistication in proposing-substrate methodology or governance thoroughness can substitute for the evidence reliability that determinism provides.

5. Mutation as the Primary Threat to the Pair

Instinct evolution per B1.10 — LLM and infrastructure upgrades arriving from upstream as undirected mutation per B1.13 — is the primary source of A1.10 threats in CKS deployments, and therefore the primary structural threat to the B1.15 + A1.10 pair. Two scenarios are architecturally distinct.

Nondeterminism introduction. An LLM update may make behavior less deterministic than under the prior model — increased output variation for identical inputs that was previously suppressed. This is a direct A1.10 contract violation: the same DNA and the same inputs must produce the same outputs. If an LLM update crosses this threshold, Action layer records accumulated under the new model become unreliable evidence. The evidence chain breaks at Step 1.

Behavioral pattern shift without nondeterminism. An LLM update may change which outputs the deployment produces for given inputs without making behavior nondeterministic. A1.10 is not violated (the new behavior is deterministic), but Action layer records from the prior era no longer accurately reflect the current DNA's behavioral patterns. Proposing substrates evaluating mixed-era records will identify patterns that conflate behavior from two different instinct-layer regimes. This threat is more subtle: records are internally consistent within each era but require era-aware evaluation across era boundaries.

Both scenarios point to the same architectural response: mutation governance instruments specified in B1.13 — verification gates per B2.06 and routing per B2.04 — must evaluate LLM updates against the determinism contract before production integration. A5.06 and A5.16 provide the operational tests for this evaluation. Integrating an LLM update without determinism verification is integrating a potential threat to the evidence chain that B1.15's action-feedback pathway depends on. Mutation governance is not only an operational concern (correctness, safety, capability) but also an evidence-integrity concern: a deployment that omits determinism verification during LLM update governance is failing to protect the action-feedback evidence chain that its continuous improvement architecture depends on.

6. Co-Presence Requirement

B1.15 without A1.10 — action-feedback governance pathway without determinism. Proposing substrates per B2.74 evaluate Action layer records that do not reliably reflect DNA-governed behavioral patterns. Proposals per B2.76 may describe patterns that are sampling artifacts of nondeterministic variation rather than stable tendencies the current DNA exhibits. Stage 1 and Stage 2 governance per B2.75 evaluate proposals without a sound evidence basis. Directed selection per B1.14 makes DNA changes that may not address the tendencies the evidence appeared to indicate. The action-feedback pathway exists as an architectural structure but is operationally degraded. This is a form of Evidence Blindness (B3.28): the pathway is present, but evidence unreliability disconnects proposals from genuine behavioral patterns, producing the same governance outcome as if the pathway were absent.

A1.10 without B1.15 — determinism without action-feedback governance pathway. Deterministic behavior is correctly established and verified by A5.06 and A5.16. Action layer records per B2.26 accumulate reliable behavioral pattern evidence with provenance metadata per A2.40. But no pathway exists through which this evidence flows into improvement proposals and governed DNA changes. The reliable evidence accumulates without governance use. Evidence Blindness (B3.28) applies here as well: reliable evidence is present but is not used for continuous, governed improvement. Determinism's governance significance — the contribution that B1.15 activates — does not materialize.

Full co-presence. A1.10 determinism produces reliable Action layer records. B1.15's action-feedback governance pathway evaluates those records through proposing substrates per B2.74, subjects proposals to Stage 1 and Stage 2 governance per B2.75, and translates authorized proposals into DNA changes through directed selection per B1.14. Each cycle produces a more refined behavioral specification and more informative evidence for the next round of action-feedback. The deployment improves in a governed, evidence-based manner that neither commitment alone can deliver.

7. Operational Integration, Detection, and Failure Modes

Operational integration. A1.10 determinism is maintained as an active property of every cell's behavior. A5.06 and A5.16 verify the contract continuously. Action layer records per B2.26 accumulate with provenance metadata per A2.40; when LLM updates occur, era-tracking in provenance supports the distinction between pre-update and post-update records in proposing-substrate evaluation. Mutation governance per B1.13 — verification gates per B2.06 and routing per B2.04 — protects A1.10 at the instinct evolution boundary before updates reach production. Proposing substrates per B2.74 evaluate determinism-verified records and produce proposals per B2.76. Stage 1 and Stage 2 governance per B2.75 receive proposals. Directed selection per B1.14 executes authorized DNA changes.

Detection. The following tests verify that the pair is functioning in co-presence:

- **A5.06 determinism test** — is cell behavior deterministic given same DNA and inputs? Failure indicates A1.10 violation and a broken evidence chain at Step 1.
- **A5.16 reproducibility test** — do operations replay deterministically? Failure confirms A1.10 violation.
- **Action-feedback pathway activity check** — is the evidence-to-proposal pathway active and producing proposals that reach governance review? Inactivity indicates B1.15 is absent from operation.
- **Pair completeness check** — are action-feedback proposals grounded in evidence verified as deterministic? Proposals based on mixed-era records without era accounting indicate a pair integration gap at the mutation boundary.
- **Mutation governance integration check** — do LLM update integration procedures include A5.06 and A5.16 verification as a prerequisite for production deployment? Absence indicates that the pair's protection against mutation-as-determinism-threat is not in place.

Failure modes. Two anti-patterns from Phase B3 are directly relevant.

Instinct-Reasoning Collapse (B3.02) — when LLM outputs bypass the reasoning layer's governance, nondeterminism can enter the deployment through the instinct layer without the verification gates per B2.06. This anti-pattern undermines A1.10 specifically at the instinct/reasoning boundary, corrupting the evidence reliability that B1.15 depends on.

Evidence Blindness (B3.28) — the absence of effective evidence use in governance. Evidence Blindness materializes both when B1.15 is absent from a deterministic deployment (reliable evidence accumulates

unused) and when A1.10 is absent from a deployment with a B1.15 pathway (the pathway evaluates unreliable evidence). Both configurations produce the same governance outcome: the deployment does not improve from its operational experience in a governed, evidence-based manner.

Conclusion

B1.15 and A1.10 form an enabling pair whose combined governance contribution exceeds what either delivers in isolation. A1.10 determinism is what makes the Action layer a reliable source of behavioral pattern evidence; B1.15's action-feedback governance pathway is what converts that evidence into governed DNA improvement. Neither delivers its governance contribution without the other: determinism without action-feedback produces reliable evidence that improves nothing; action-feedback without determinism evaluates noise that cannot be distinguished from genuine behavioral patterns. Mutation governance instruments protecting A1.10 during instinct evolution directly protect the evidence chain that B1.15 depends on. Full co-presence of A1.10 and B1.15 — with mutation governance maintaining the pair's integrity at the instinct evolution boundary — is the architectural condition under which a CKS deployment learns systematically from its operational experience and improves in a governed, evidence-based manner.

Source papers

Li, W. (2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems*. April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance*. April 2026. ORCID: 0009-0004-8065-3235.

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